GALAXY	GALAXY TRANSMISSIONS PVT. LTD. *CONDUCTOR DIVISON* *WORKS : Plot Noi N-3, MIDC Kupwad Block, Sangli - 416436.*
CUSTOMER	**M/s TRANSMISSION CORPORATION OF ANDHRA PRADESH LTD (APTRANSCO)**
CONTRACTOR:	**M/s P CHANDRA REDDY & COMPANY**
PROJECT :	**Tender Notice No:46/2023-24:- Shifting and commissioning of three phase double circuit 132 KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132 KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO**

GTP / DATA SHEET FOR AAA PANTHER CONDUCTOR

SR NO	DESCRIPTION	UNIT	PARTICULARS
1	Name & Address of Manufacturer	-	**GALAXY TRANSMISSIONS PVT. LTD.**
2	Code Name & Size	-	AAAC PANTHER (37AL/2.88MM)
2.1	Quality of material & standard to which conform.	-	IS 398 PART IV :1994
3	**Particulars of Raw Materials :**		
3.1	Aluminium	-	
	(a) Aluminium wire Rod		Galaxy Aluminium LLP
	(b) Aluminium Ingot		BALCO/VEDANTA/NALCO/HINDALCO
3.2	Aluminium Alloy Wire	-	
	Grade - 6201 Chemical Composition		
	(a) Si	%	0.50 to 0.90
	(b) Mg	%	0.60 to 0.90
	(c) Fe	%	0.50 Max
	(d) Cu	%	0.10 Max
	(e) Mn	%	0.03 Max
	(f) Cr	%	0.03 Max
	(g) Zn	%	0.10 Max
	(h) B	%	0.06 Max
	(i) Other Elements (Each)	%	0.03 Max
	(i) Other Elements (Total)	%	0.10 Max
	(k) Al	%	Remainder
4	**ALUMINIUM STRANDS AFTER STRANDING**		
4.1	Diameter	-	
	(a) Nominal	mm	2.88
	(b) Maximum	mm	2.91
	(c) Minimum	mm	2.85
4.2	Minimum Breaking load of strand.	-	
	(a) Before Stranding	kN	2.02
	(b) After Stranding	kN	1.92
4.3	% Elongation on 250mm GL	%	4.0 Min
5	**AAA CONDUCTOR**	-	
5.1	Outer Diameter of Conductor	mm	20.16 ±1%
5.2	Cross Sectional area of		
	a) Whole Conductor	mm^2	241
	b) Each Strand	mm^2	6.514
5.3	Minimum % of Aluminium in Alloy	%	97.31
	Number Of Strands		
	a) Centre wire		1

APTRANSCO Approved Boundary Only.

GALAXY	GALAXY TRANSMISSIONS PVT. LTD. *CONDUCTOR DIVISON* *WORKS : Plot Noi N-3, MIDC Kupwad Block, Sangli - 416436.*
CUSTOMER	**M/s TRANSMISSION CORPORATION OF ANDHRA PRADESH LTD (APTRANSCO)**
CONTRACTOR:	**M/s P CHANDRA REDDY & COMPANY**
PROJECT :	**Tender Notice No:46/2023-24:- Shifting and commissioning of three phase double circuit 132 KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132 KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO**

GTP / DATA SHEET FOR AAA PANTHER CONDUCTOR

SR NO	DESCRIPTION	UNIT	PARTICULARS
5.4	b) First Layer	Nos	6
	c) Second Layer		12
	d) Third Layer		18
5.5	Weight of	kg. / km.	
	a) Whole Conductor		664 ± 2%
	b) Aluminium Alloy Strand at Nominal dia.		17.59
5.6	UTS of Conductor	kN	67.49 (Minimum)
5.7	Lay ratio of conductor	-	Max Min
	(a) 07 Wire aluminium layer		14 12
	(b) 12 Wire Aluminium Alloy layer		13 11
	(c) 18 Wire Aluminium Alloy layer		12 10
5.8	Calculated DC resistance at 20 deg C		
	a) Whole Conductor	Ohm/km	0.1417 (Maximum)
	b) Aluminium Alloy Strand	Ohm/km	4.989 (Maximum)
5.9	a) Current Carrying Capacity of conductor at Ambient temperature 40°C 45°C	Amps	470A 563 A @ 85°C
	b) Temperature rise above ambient temp.	°C	45
6	Resistivity	Ohms.sqmm/Mtr	0.03284 Max.
6.1	Co-efficient of linear expansion	per °C	23.0×10^{-6}
	Modulus of elasticity of whole conductor		0.5814×10^{6}
6.2	Standard length of Conductor	M	1500
6.3	Tolerance on standard length of Conductor	%	± 5%
6.4	Number of standard length in one reel	Nos	1
	Weight of Conductor in one reel	Kg	996 Approx
	Weight of the Drum	Kg	150±5%
	Gross weight of the reel including weight of the Conductor	Kg	1146 Approx
	Drum Details (Dimension) in mm	FxBxT	1295x500x750
6.5	Direction of lay for outside layer	NA	Right Hand
6.6	Linear mass of the Conductor	-	-
	(a) Standard	Kg/km	664
	(b) Minimum	Kg/km	651
	(c) Maximum	Kg/km	677

Medium grade Kraft paper shall be used in between the layers of the conductor. After reeling the conductor, the exposed surface of the outer layer of the conductor shall be wrapped with thin polythene sheet across the flanges.

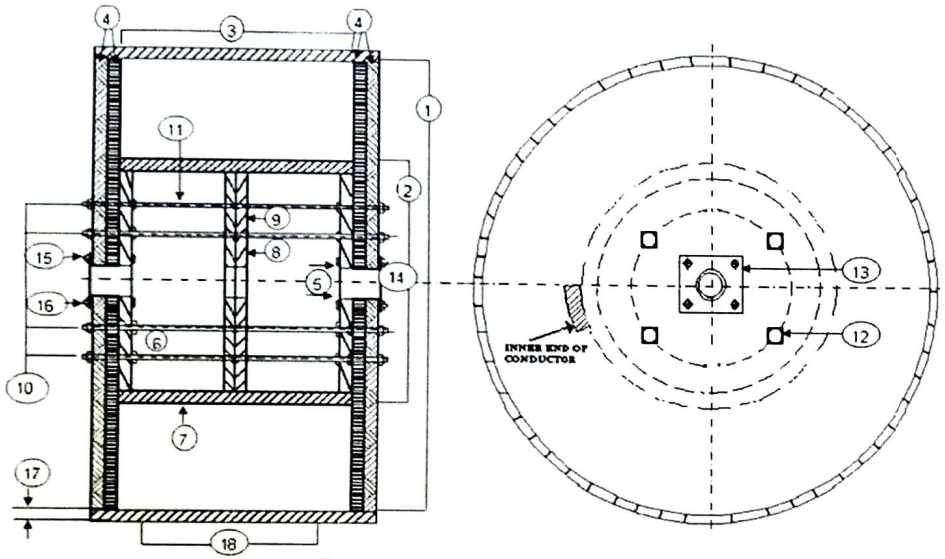
For Galaxy Transmissions Pvt. Ltd.,

CHECKED

DEE-1 DESIGNS

DEE-2 DESIGNS

EE DESIGNS

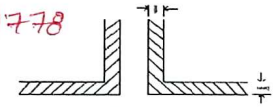


Sr	DRUM COMPONENTS	MARK No	SIZE / QTY
1	FLANGE DIAMETER	1	1295
2	BARREL DIAMETER INCLUDING LAGGING	2	500
3	TRAVERSE	3	750
4	FLANGE THICKNESS	4	2 X 28
5	BORE DIAMETER	5	110
6	THICKNESS OF BARREL END SUPPORT	6	28
7	THICKNESS OF BARREL END LAGGING	7	28
8	THICKNESS OF CENTER DISC	8	28 X 2
9	SIZE Of CENTER DISC	9	450 X 28 X 2
10	NO OF TIE RODS	10	4
11	DIAMETER OF TIE ROD	11	16
12	SIZE OF SQUARE WASHER	12	50X50X4
13	SIZE Of SPINDLE PLATE	13	150X150 X4
14	DIAMETER OF SPINDLE PLATE HOLE	14	100
15	NO. OF SPINDLE PLATE BOLT	15	4
16	SPINDLE PLATE BOLTS DIAMETER	16	12
17	THICKNESS OF EXTERNAL PP SHEET	17	3
18	NO. OF BINDER OVER EXTERNAL PP SHEET	18	2

the Drums shall be IS: 1778

BARREL BATTEN - CROSS SECTION





BORE PIPE DETAIL

CHECKED

DEE-1 DESIGNS

DEE-2 DESIGNS

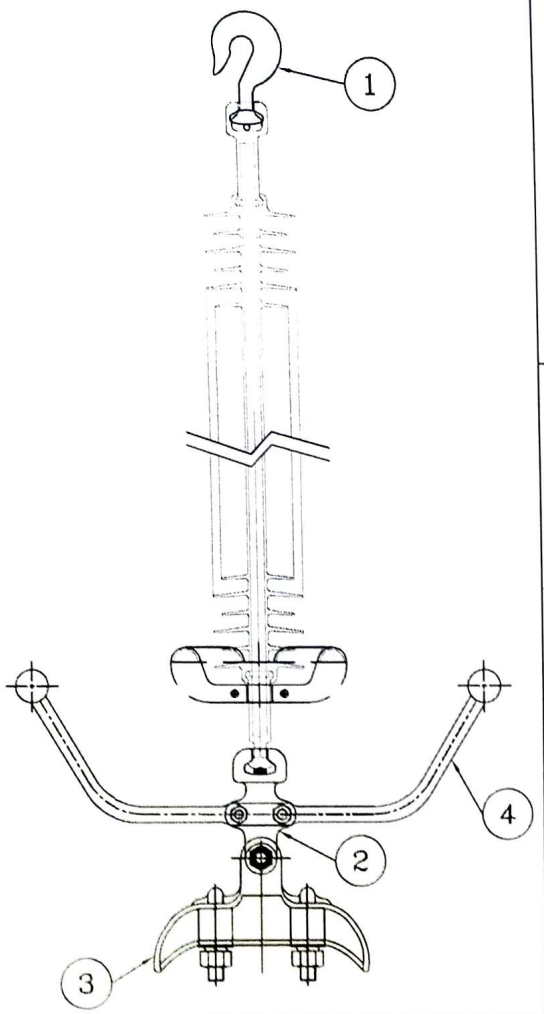
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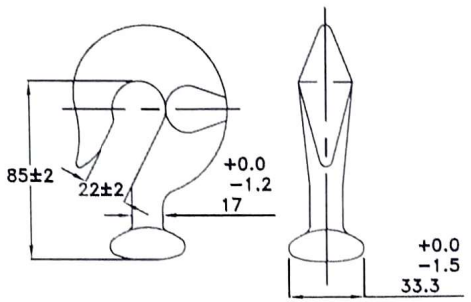
NOTE :	
1.	ALL DIMENSIONS ARE IN M.M
2.	THE STANDARD LENGTH OF THE AAAC PANTHER AL37/2.88 MM SIZE OF CONDUCTOR IS 1500 MTRS WITH TOLERANCE OF +/- 5%
3.	ONE LENGTH (1500 x 1 = 1500 MTR.)OF THE CONDUCTOR SHALL BE WOUND ON EVERY DRUM.
4.	TOLERANCE ON DIMENSIONS OF WOOD IS ONLY +/- 3 MM
5.	NET WEIGHT OF CONDUCTOR 996 KGS (APPROX)
6.	TARE WEIGHT OF DRUM 150 KGS (APPROX)
7.	GROSS WEIGHT OF DRUM WITH CONDUCTOR 1146 KGS (APPROX)
8	PP SHEET SHALL BE USED FOR DRUM PACKING

Marking on Drum : 1) Customer Name 2) Mfg. Name. 3) Drum No 4) Size of Conductor 5) Length of Conductor 6) Number of Length 7) Gross/Net/Tare Wt. of Conductor 8) Arrow mark for Unwinding

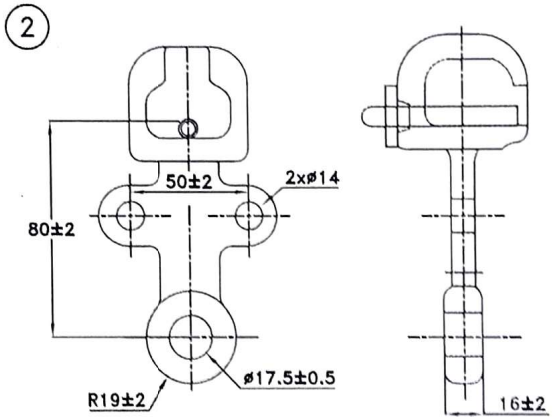
GALAXY	GALAXY TRANSMISSIONS PVT. LTD. UNIT 1 PLOT NO N-3, MIDC, KUPWAD BLOCK, SANGLI.
UNIT 1	PLOT NO N-3, MIDC, KUPWAD BLOCK, SANGLI.
DRG. NO.	GTPL/QA/2023-24/AAAC PANTHER/DD-51"/020324 REV 00
TITLE	WOODEN DRUM DRAWING FOR AAAC PANTHER CONDUCTOR
CUSTOMER	M/s TRANSMISSION CORP OF ANDHRA PRADESH LTD (APTRANSCO)
CONTRACTOR:	M/s P CHANDRA REDDY & COMPANY
PROJECT	Tender Notice No:46/2023-24:- Shifting and commissioning of three phase double circuit 132 KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132 KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO
CODE NAME	AAAC PANTHER (37AL/2.88MM)
DRWN BY	NBP
APPROVED BY	

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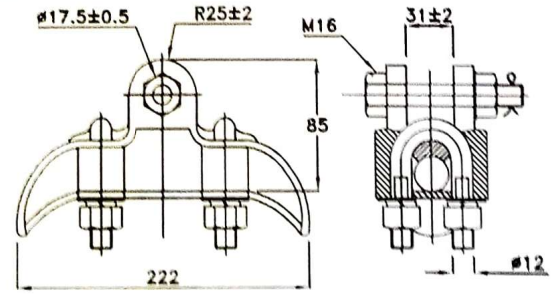




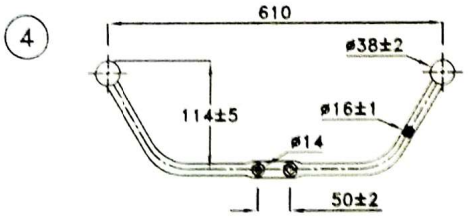
1 BALL HOOK



2 HH SOCKET EYE



3 SUSPENSION CLAMP



4 ARCHING HORN

-: TECHNICAL DATA :-

(1) ALL DIMENSION ARE IN mm.
(2) GENERAL TOLERANCE :±3% WHERE NOT SPECIFIED.
(3) ALL FERROUS PARTS ARE HDG AS PER IS:2629 & IS:2633.
(4) BALL & SOCKET SIZE = 16 mm 'B' AS PER IS:2486-II.
(5) UTS OF HW : 50% OF CONDUCTOR UTS. 21.0 KN
(6) SLIP STRENGTH : 8% TO 15% OF CONDUCTOR UTS.

CHECKED

DEE-1 DESIGNS — DEE-2 DESIGNS — EE DESIGNS

AUMNI TRANSMISSION INDUSTRY PVT. LTD. VADODARA

BILL OF MATERIAL

SR. NO.	DESCRIPTION	SPECIFICATION	QTY.
6	SPLIT PIN	STAINLSS STEEL	1
5	R CLIP	STAINLSS STEEL	1
4	ARCHING HORN	MILD STEEL – H.D.G.	1
3	SUSPENSION CLAMP	ALU. ALLOY-LM 6	1
2	HH SOCKET EYE	FORGED STEEL – H.D.G.	1
1	BALL HOOK	FORGED STEEL – H.D.G.	1

	PCR/VPA/PO/004, Date:01-03-2024
End Customer	M/s.Transmission Corporation of A.P Ltd. (APTRANSCO)
EPC Contractor	M/s.P.ChandraReddy & Co., Tirupati
Project Name	Tender Notice No:46/2023-24:- Shifting and commissioning of three phase double circuit 132KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO Visakhapatnam.
APTRANSCO PO. No. :	Lr.No.CE/VSP Zone/O&M/DEE-1/AEE4/F.DOC/D.No.385/24, Date:05-02-2024.

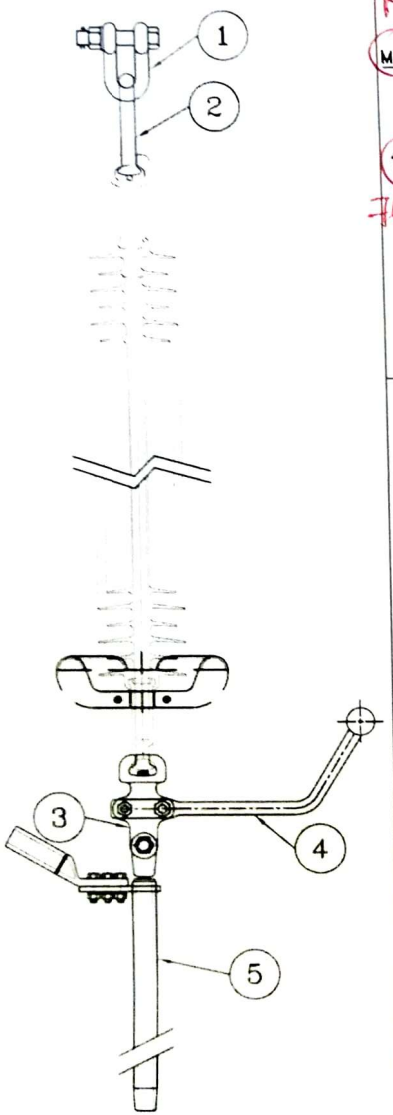
TITLE: SINGLE SUSPENSION PILOT HARDWARE FOR AAAC PANTHER CONDUCTOR

AUMNI TRANSMISSION INDUSTRY PVT.LTD.
PLOT NO.E-76 MANJUSAR GIDC, TAL: SAVLI,
DIST:VADODARA,GUJARAT

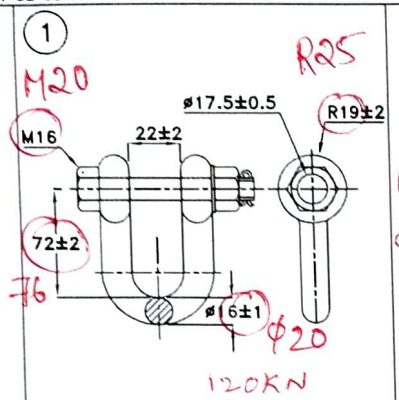
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DRAWN	HARSHAD.M	REVISION STATUS			
CHECKED	M.A.ZALA	NO.	DATE	DESCRIPTION	APP.
APPROVED	NS				
DATE					

\\aumni\TL_DESIGN\CAD Works\Govt. Parties\AP TRANCO\132 KV Panther\HW

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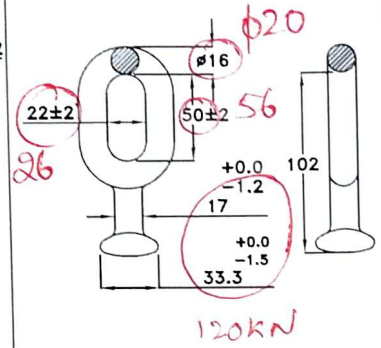


1



ANCHOR SHACKLE

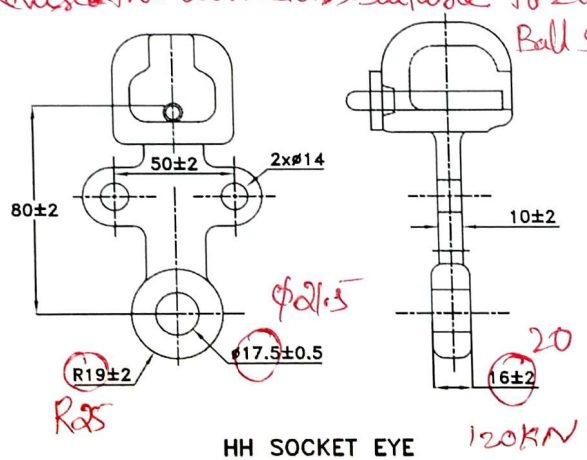
2



BALL LINK

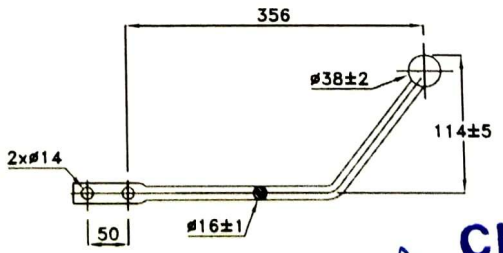
3

Revise the dimensions suitable to 20mm Ball Size.



HH SOCKET EYE

4



ARCHING HORN

TECHNICAL DATA :-

(1) ALL DIMENSION ARE IN mm. 120mm
(2) GENERAL TOLERANCE :±3% WHERE NOT SPECIFIED.
(3) ALL FERROUS PARTS ARE HDG AS PER IS:2629 & IS:2633.
(4) BALL & SOCKET SIZE = 16 mm 'B' AS PER IS:2486-II.
(5) UTS : 100% OF CONDUCTOR UTS. = 120KN
(6) SLIP STRENGTH : 95% OF CONDUCTOR UTS.

AUMNI TRANSMISSION IND. PVT. LTD. VADODARA

BILL OF MATERIAL

SR. NO.	DESCRIPTION	SPECIFICATION	QTY.
7	SPLIT PIN	STAINLSS STEEL	2
6	R CLIP	STAINLSS STEEL	1
5	COMPRESSION DEAD END ASSEMBLY	ALU. ALLOY- 63401 IS:5082-1998 & FORGED STEEL- H.D.G CL-II, IS:2004	1
4	ARCHING HORN	MILD STEEL - H.D.G.	1
3	HH SOCKET EYE	FORGED STEEL - H.D.G.	1
2	BALL LINK	FORGED STEEL - H.D.G.	1
1	ANCHOR SHACKLE	FORGED STEEL - H.D.G.	1

PCR/VPA/PO/004, Date:01-03-2024	
End Customer	M/s.Transmission Corporation of A.P Ltd. (APTRANSCO)
EPC Contractor	M/s.P.ChandraReddy & Co., Tirupati
Project Name	Tender Notice No:46/2023-24:- Shifting and commissioning of three phase double circuit 132KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO Visakhapatnam.
APTRANSCO PO. No. :	Lr.No.CE/VSP Zone/O&M/DEE-1/AEE4/F.DOC/D.No.365/24, Date:05-02-2024.

TITLE: SINGLE TENSION HARDWARE FOR AAAC PANTHER CONDUCTOR

20.16

AUMNI TRANSMISSION INDUSTRY PVT.LTD.
PLOT NO.E-76 MANJUSAR GIDC, TAL: SAVLI,
DIST:VADODARA,GUJARAT

SCALE	NTS	REVISION STATUS			
DRAWN	HARSHAD.M	DRG.NO. : ATIPL-ST-AAAC-P-01			
CHECKED	M.C.ZALA	NO.	DATE	DESCRIPTION	APP.
APPROVED	NS				
DATE					

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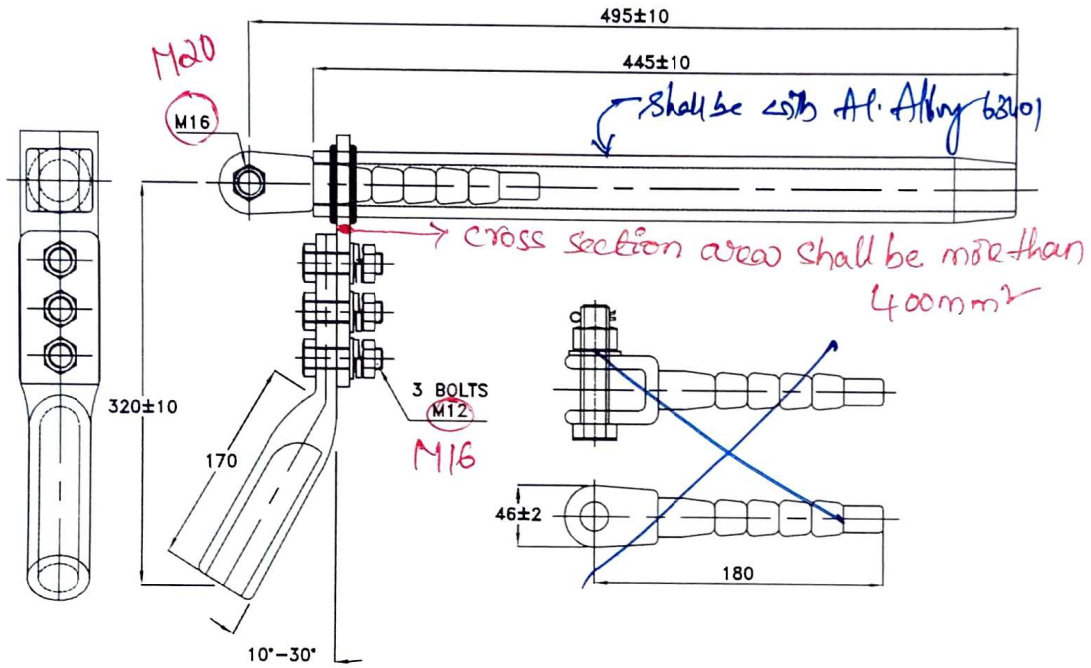
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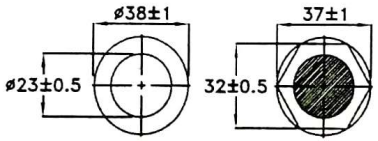
\\aumni\TL_DESIGN\CAD Works\Govt. Parties\AP TRANCO\132 KV Panther\HW

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(5)

AAAC panther 20 - 16 ϕ dia





BEFORE COMP. AFTER COMP.

ALUMINIUM SLEEVE

COMPRESSION DEAD END

TECHNICAL DATA :-

(1) ALL DIMENSION ARE IN mm. 120 MM
(2) GENERAL TOLERANCE :±3% WHERE NOT SPECIFIED.
(3) ALL FERROUS PARTS ARE HDG AS PER IS:2629 & IS:2633.
(4) BALL & SOCKET SIZE = 16 mm 'B' AS PER IS:2486–II.
(5) UTS : 100% OF CONDUCTOR UTS. — 120 KN
(6) SLIP STRENGTH : 95% OF CONDUCTOR UTS.

AUMNI TRANSMISSION IND. PVT. VADODARA

BILL OF MATERIAL

SR. NO.	DESCRIPTION	SPECIFICATION	QTY.
7	SPLIT PIN	STAINLSS STEEL	2
6	R CLIP	STAINLSS STEEL	1
5	COMPRESSION DEAD END ASSEMBLY	ALU. ALLOY– 63401 IS:5082–1998 & FORGED STEEL– H.D.G CL–II, IS:2004	1
4	ARCHING HORN	MILD STEEL – H.D.G.	1
3	HH SOCKET EYE	FORGED STEEL – H.D.G.	1
2	BALL LINK	FORGED STEEL – H.D.G.	1
1	ANCHOR SHACKLE	FORGED STEEL – H.D.G.	1

PCR/VPA/PO/004, Date:01–03–2024

End Customer	M/s.Transmission Corporation of A.P Ltd. (APTRANSCO)
EPC Contractor	M/s.P.ChandraReddy & Co., Tirupati
Project Name	Tender Notice No:46/2023–24:– Shifting and commissioning of three phase double circuit 132KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO Visakhapatnam.
APTRANSCO PO. No. :	Lr.No.CE/VSP Zone/O&M/DEE–1/AEE4/F.DOC/D.No.365/24, Date:05–02–2024.

TITLE: SINGLE TENSION HARDWARE FOR AAAC PANTHER CONDUCTOR

AUMNI TRANSMISSION INDUSTRY PVT.LTD.
PLOT NO.E-76 MANJUSAR GIDC, TAL: SAVLI,
DIST:VADODARA,GUJARAT

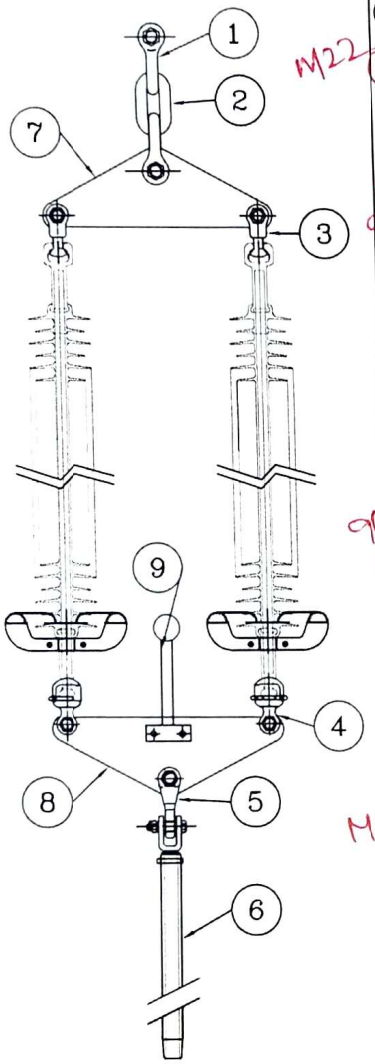
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DRAWN	HARSHAD.M	DRG.NO. : ATIPL–ST–AAAC–P–02				
CHECKED	M.A.ZALA	NO.	DATE	DESCRIPTION	APP.	
APPROVED	NS					
DATE						

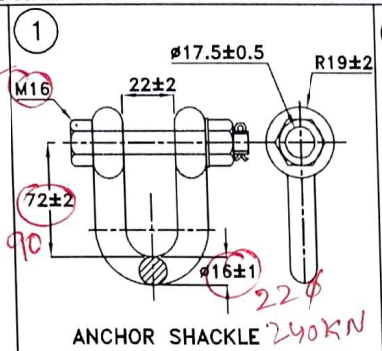
\\aumni\TL_DESIGN\CAD Works\Govt. Parties\AP TRANCO\132 KV Panther\HW

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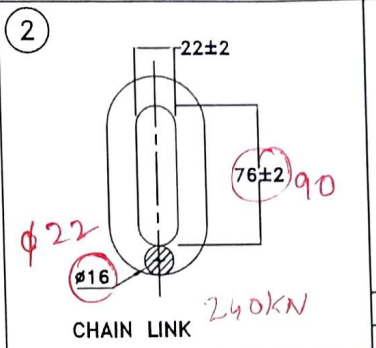
DEE-1 DESIGNS | DEE-2 DESIGNS | EE DESIGNS

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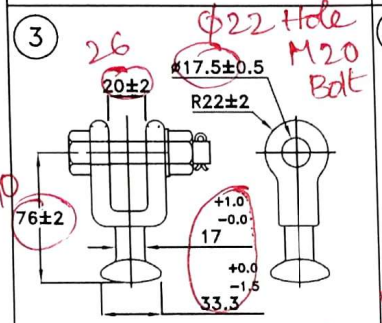




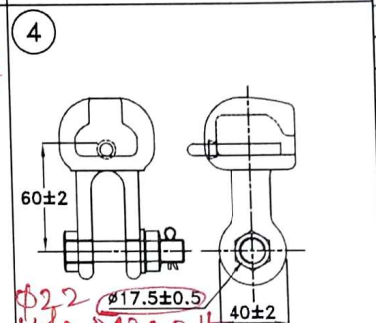
ANCHOR SHACKLE 240KN



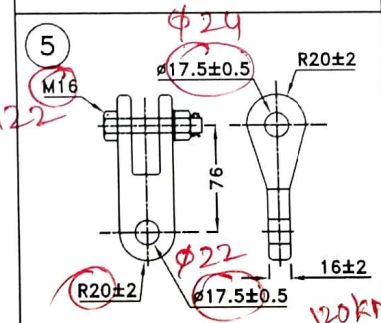
CHAIN LINK 240KN



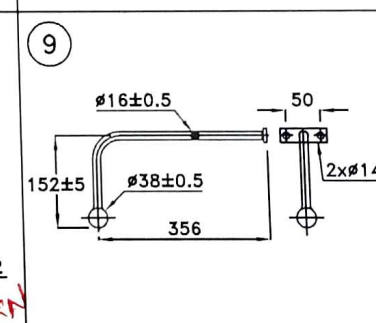
BALL CLEVIS 120KN



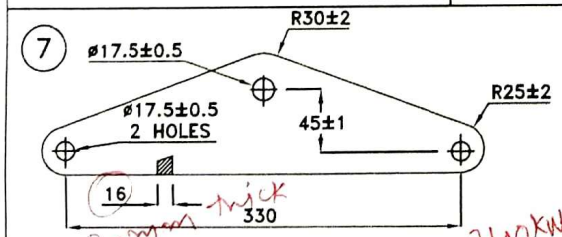
SOCKET CLEVIS 120KN.



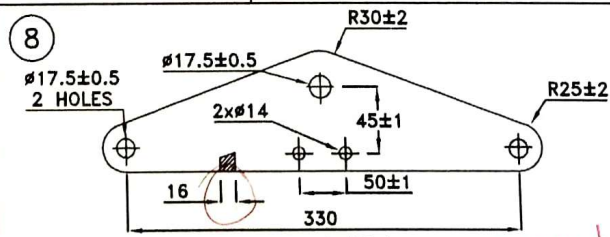
CLEVIS CLEVIS 120KN



ARCHING HORN



YOKE PLATE TS 240KN



YOKE PLATE LS 240KN

TECHNICAL DATA :-

(1) ALL DIMENSION ARE IN mm.
(2) GENERAL TOLERANCE :±3% WHERE NOT SPECIFIED.
(3) ALL FERROUS PARTS ARE HDG AS PER IS:2629 & IS:2633.
(4) BALL & SOCKET SIZE = 16 mm 'B' AS PER IS:2486-II.
(5) UTS : 100% OF CONDUCTOR UTS.
(6) SLIP STRENGTH : 95% OF CONDUCTOR UTS.

BILL OF MATERIAL

SR. NO.	DESCRIPTION	SPECIFICATION	QTY.
11	SPLIT PIN	STAINLSS STEEL	8
10	R CLIP	STAINLSS STEEL	2
9	ARCHING HORN	MILD STEEL – H.D.G.	1
8	YOKE PALTE LS	MILD STEEL – H.D.G.	1
7	YOKE PALTE TS	MILD STEEL – H.D.G.	1
6	COMPRESSION DEAD END ASSEMBLY	ALU. ALLOY– 63401 IS:5082–1998 & FORGED STEEL– H.D.G CL–II, IS:2004	1
5	CLEVIS EYE	FORGED STEEL – H.D.G.	1
4	SOCKET CLEVIS	FORGED STEEL – H.D.G.	2
3	BALL CLEVIS	FORGED STEEL – H.D.G.	2
2	CHAI LINK	FORGED STEEL – H.D.G.	1
1	ANCHOR SHACKLE	FORGED STEEL – H.D.G.	2

PCR/VPA/PO/004, Date:01-03-2024

End Customer	M/s.Transmission Corporation of A.P Ltd. (APTRANSCO)
EPC Contractor	M/s.P.ChandraReddy & Co., Tirupati
Project Name	Tender Notice No:46/2023-24:- Shifting and commissioning of three phase double circuit 132KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO Visakhapatnam.
APTRANSCO PO. No. :	Lr.No.CE/VSP Zone/O&M/DEE-1/AEE4/F.DOC/D.No.365/24, Date:05-02-2024.

TITLE: DOUBLE TENSION HARDWARE FOR AAAC PANTHER CONDUCTOR

AUMNI TRANSMISSION INDUSTRY PVT.LTD.
PLOT NO.E-76 MANJUSAR GIDC, TAL: SAVLI,
DIST:VADODARA,GUJARAT

SCALE	NTS	REVISION STATUS				
DRAWN	HARSHAD.M	DRG.NO. : ATIPL-DT-AAAC-P-01				
CHECKED	M.A.ZALA	NO.	DATE	DESCRIPTION		APP.
APPROVED	NS					
DATE						

CHECKED

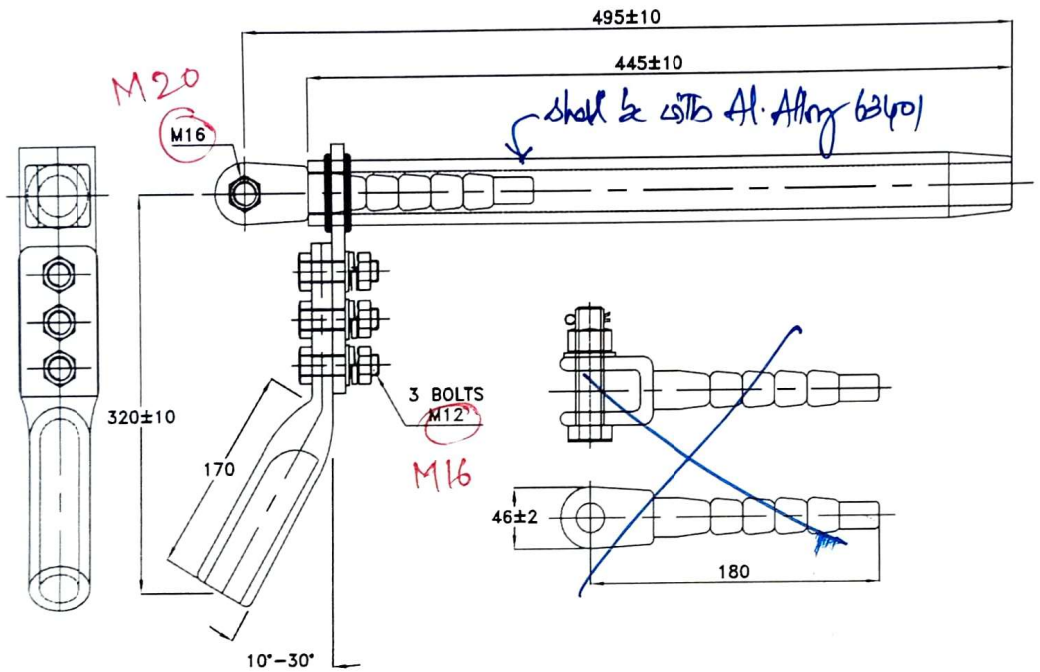
DEE-1 DESIGNS | DEE-2 DESIGNS | EE DESIGNS

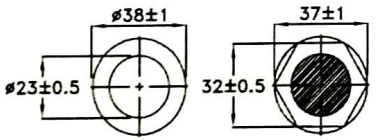
\\aumni\TL_DESIGN\CAD Works\Govt. Parties\AP TRANCO\132 KV Panther\HW

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6

AAAC panther 20.16 dia





ALUMINIUM SLEEVE

COMPRESSION DEAD END

TECHNICAL DATA :-

(1) ALL DIMENSION ARE IN mm.
(2) GENERAL TOLERANCE :±3% WHERE NOT SPECIFIED.
(3) ALL FERROUS PARTS ARE HDG AS PER IS:2629 & IS:2633.
(4) BALL & SOCKET SIZE = 16 mm 'B' AS PER IS:2486–II.
(5) UTS : 100% OF CONDUCTOR UTS. — 120kN
(6) SLIP STRENGTH : 95% OF CONDUCTOR UTS.

720mm

AUMNI TRANSMISSION IND. PVT. LTD. VADODARA

BILL OF MATERIAL

SR. NO.	DESCRIPTION	SPECIFICATION	QTY.
11	SPLIT PIN	STAINLSS STEEL	8
10	R CLIP	STAINLSS STEEL	2
9	ARCHING HORN	MILD STEEL – H.D.G.	1
8	YOKE PALTE LS	MILD STEEL – H.D.G.	1
7	YOKE PALTE TS	MILD STEEL – H.D.G.	1
6	COMPRESSION DEAD END ASSEMBLY	ALU. ALLOY– 63401 IS:5082–1998 & FORGED STEEL– H.D.G CL–II, IS:2004	1
5	CLEVIS EYE	FORGED STEEL – H.D.G.	1
4	SOCKET CLEVIS	FORGED STEEL – H.D.G.	2
3	BALL CLEVIS	FORGED STEEL – H.D.G.	2
2	CHAI LINK	FORGED STEEL – H.D.G.	1
1	ANCHOR SHACKLE	FORGED STEEL – H.D.G.	2

PCR/VPA/PO/004, Date:01–03–2024

End Customer	M/s.Transmission Corporation of A.P Ltd. (APTRANSCO)
EPC Contractor	M/s.P.ChandraReddy & Co., Tirupati
Project Name	Tender Notice No:46/2023–24:– Shifting and commissioning of three phase double circuit 132KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO Visakhapatnam.
APTRANSCO PO. No. :	Lr.No.CE/VSP Zone/O&M/DEE–1/AEE4/F.DOC/D.No.365/24, Date:05–02–2024.

TITLE: DOUBLE TENSION HARDWARE FOR AAAC PANTHER CONDUCTOR 20.16 ϕ dia

AUMNI TRANSMISSION INDUSTRY PVT.LTD.
PLOT NO.E-76 MANJUSAR GIDC, TAL: SAVLI,
DIST:VADODARA,GUJARAT

SCALE	NTS	REVISION STATUS				
DRAWN	HARSHAD.M	DRG.NO. : ATIPL–DT–AAAC–P–02				
CHECKED	M.A.ZALA	NO.	DATE	DESCRIPTION	APP.	
APPROVED	NS					
DATE						

CHECKED

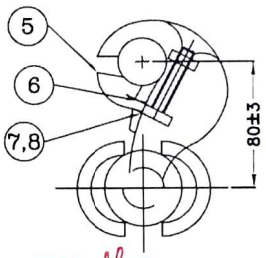
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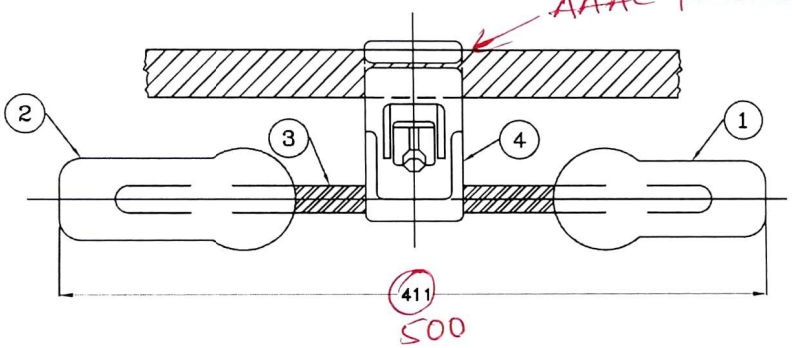
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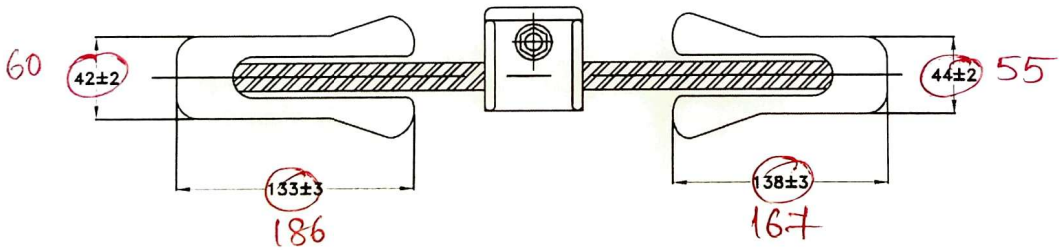
EE DESIGNS

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CHECKED

DEE-1 DESIGNS

DEE-2 DESIGNS

EE DESIGNS

TECHNICAL DATA:–

1) ALL DIMENSION ARE IN MM.
2) GENERAL.TOLERANCE ±5% UNLESS OTHERWISE SPECIFIED.
3) FASTENERS HOT DIP GALVANIZING & SPRING WASHER ELECTRO GALVANIZED.
4) TOTAL WEIGHT OF DAMPER ASSEMBLY : 2.4 KG (APPROX)
5) MESSENGER CABLE SIZE: 19/2.0 & 135 Kg/ mm2.
6) WEIGHT IS INDICATIVE AND NOT MANDATORY.
7) CLAMP TORQUE : 4 KGM.

Damping frequency shall be ensured during Testing

AUMNI TRANSMISSION INDUSTRY PVT. LTD. VADODARA

BILL OF MATERIAL

SR.NO.	DESCRIPTION	MATERIAL	QTY.
1.	DAMPER WEIGHT–B	CAST IRON–HDG	1
2.	DAMPER WEIGHT–S	CAST IRON–HDG	1
3.	MESSENGER CABLE	19 STRANDS HIGH T.S.SPECIAL STEEL–H.D.G.	1
4.	CLAMP BODY	ALUMINIUM ALLOY LM–6	1
5.	KEEPER	ALUMINIUM ALLOY LM–6	1
6.	PLAIN WASHER	MILD STEEL–HDG	1
7.	HEX BOLT & NUT.	MILD STEEL–HDG	1
8.	SPRING WASHER	SPRING STEEL–EG	1

PCR/VPA/PO/004, Date:01–03–2024

End Customer	M/s.Transmission Corporation of A.P Ltd. (APTRANSCO)
EPC Contractor	M/s.P.ChandraReddy & Co., Tirupati
Project Name	Tender Notice No:46/2023–24:– Shifting and commissioning of three phase double circuit 132KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO Visakhapatnam.
APTRANSCO PO. No. :	Lr.No.CE/VSP Zone/O&M/DEE–1/AEE4/F.DOC/D.No.365/24, Date:05–02–2024.

TITLE '4–R' DAMPER FOR AAAC PANTHER CONDUCTOR

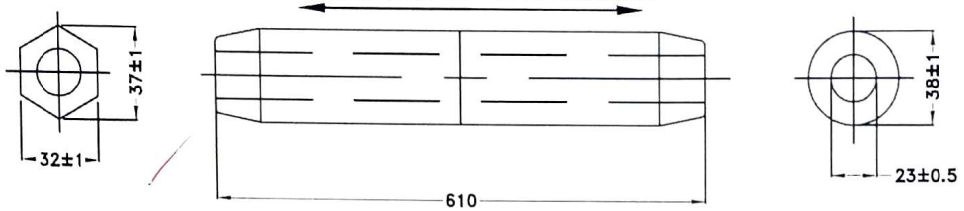
AUMNI TRANSMISSION INDUSTRY PVT.LTD.
PLOT NO.E-76 MANJUSAR GIDC, TAL: SAVLI,
DIST:VADODARA,GUJARAT

SCALE	NTS	REVISION STATUS				
DRAWN	HARSHAD.M	DRG.NO. : ATIPL–VD–AAAC–P				
CHECKED	M.A.ZALA	NO.	DATE	DESCRIPTION	APP.	
APPROVED	NS					
DATE						

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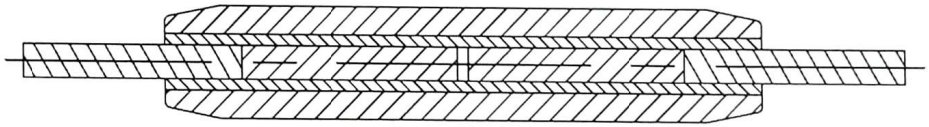
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DIRECTION OF COMPRESSION



AFTER COMPRN. BEFORE COMPRN.

ALUMINIUM SLEEVE MAT. :- AL.ALLOY 63401



ASSEMBLY

TECHNICAL DATA :-

(1) ALL DIMENSION ARE IN mm.
(2) GENERAL TOLERANCE :±3% WHERE NOT SPECIFIED.
(3) MIN. BREAKING STRENGHT 95% OF UTS OF CONDUCTOR.
(4) COMPRESSION PRESSURE : 150 TONS.
(5) MATERIAL- ALU. ALLOY- 63401 IS:5082-1998.

CHECKED

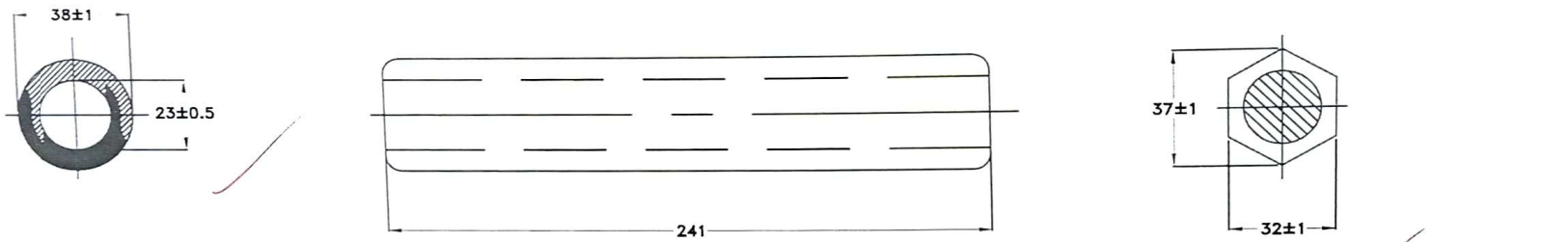
DEE-1 DESIGNS | DEE-2 DESIGNS | EE DESIGNS

PCR/VPA/PO/004, Date:01-03-2024					
End Customer	M/s.Transmission Corporation of A.P Ltd. (APTRANSCO)				
EPC Contractor	M/s.P.ChandraReddy & Co., Tirupati				
Project Name	Tender Notice No:46/2023-24:- Shifting and commissioning of three phase double circuit 132KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO Visakhapatnam.				
APTRANSCO PO. No. :	Lr.No.CE/VSP Zone/O&M/DEE-1/AEE4/F.DOC/D.No.365/24, Date:05-02-2024.				
TITLE	MID SPAN JOINT AAAC PANTHER CONDUCTOR				
AUMNI TRANSMISSION INDUSTRY PVT.LTD. PLOT NO.E-76 MANJUSAR GIDC, TAL: SAVLI, DIST:VADODARA,GUJARAT					
SCALE	NTS	REVISION STATUS			
DRAWN	HARSHAD.M	DRG.NO. : ATIPL-MSJ-AAAC-P			
CHECKED	M.A.ZALA	NO.	DATE	DESCRIPTION	APP.
APPROVED	NS				
DATE					

AUMNI TRANSMISSION IND. PVT. LTD. VADODARA

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BEFORE COMPRESSION

AFTER COMPRESSION

AUMNI TRANSMISSION IND. PVT. VADODARA

PCR/VPA/PO/004, Date:01-03-2024

End Customer	M/s.Transmission Corporation of A.P Ltd. (APTRANSCO)
EPC Contractor	M/s.P.ChandraReddy & Co., Tirupati
Project Name	Tender Notice No:46/2023-24:- Shifting and commissioning of three phase double circuit 132KV NCR Transmission line in connection with proposed rerouting of the 9 nos of 132KV tower lines of 132KV HNPCL, APCL, RCL, ESSR, Gajuwaka lines which are crossing the main & loop railway tracks of western sector areas near VPA lands in the jurisdiction of Chief Engineer /APTRANSCO Visakhapatnam.
APTRANSCO PO. No. :	Lr.No.CE/VSP Zone/O&M/DEE-1/AEE4/F.DOC/D.No.365/24, Date:05-02-2024.
TITLE	REPAIR SLEEVE AAAC PANTHER CONDUCTOR

TECHNICAL DATA :-

(1) ALL DIMENSION ARE IN mm.
(2) GENERAL TOLERANCE :±3% WHERE NOT SPECIFIED.
(3) MIN. BREAKING STRENGHT 95% OF UTS OF CONDUCTOR.
(4) COMPRESSION PRESSURE : 100 TONS.
(5) MATERIAL- ALUMINUM AS PER IS:733, Gr.-19500.

CHECKED

DEE-1 DESIGNS

DEE-2 DESIGNS

EE DESIGNS

AUMNI TRANSMISSION INDUSTRY PVT.LTD.
PLOT NO.E-76 MANJUSAR GIDC, TAL: SAVLI,
DIST:VADODARA,GUJARAT

SCALE	NTS	REVISION STATUS			
DRAWN	HARSHAD.M	DRG.NO. : ATIPL-RS-AAAC-P			
CHECKED	M.A.ZALA	NO.	DATE	DESCRIPTION	APP.
APPROVED	NS				
DATE					

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